



Tying it together

Scientific models + machine learning — and where it's heading

Keep the science — let neural nets fill the gaps

We never threw the mechanistic model away. We let **neural networks** do two jobs **inside** it:

DISCOVER THE DYNAMICS

Where the mechanism is unknown, a neural net learns the missing term(s) — **UDEs** and **Neural ODEs**. A dial from hand-written mechanism to fully data-driven.

POWER THE RANDOM EFFECTS

The **individual** layer — how each subject differs from the population, and how the population itself varies.

Almost everything else today was a consequence of these two.

Individualizable functions — or a distribution you can reshape

INDIVIDUALIZABLE FUNCTIONS

A random effect turns the structural model into a **per-subject** function — individualization, without fitting a separate model for everyone.

TRANSFORMATIONS OF THE BSV DISTRIBUTION

The random-effect prior **is** the between-subject-variability distribution — and we can **reshape** it: condition it on covariates (**augment**), **join** models with a learned joint prior, pour in **embeddings**.

WHY IT MATTERED

Seeing the random effects as a **latent distribution** — the generative view — is exactly what let us borrow ideas from machine learning.

A lot follows from those two moves

- 1 **Discover** prognostic factors and covariate relationships **from data**, instead of pre-specifying them.
- 2 Bring **complex data** — text, images — in as covariates through **embeddings**.
- 3 **Join** independently-built endpoint models so information **flows across** them.
- 4 **Individualized** predictions from sparse data — and richer, mechanism-respecting **virtual populations**.

The next leaps

FIT AT A DIFFERENT SCALE

The bottleneck is **fitting**. The biggest levers I see are **Variational EM (VEM)** and **reverse-mode** differentiation — they change how the fit **scales**, beyond what FOCE reaches, to **large joint and neural** models.

COMPLEX DATA → LARGE JOINT NLME

Turn images and text into embeddings, **track them over time**, and build **large joint NLME models** relating those trajectories to both the **drug (conditional)** and the **reportable clinical endpoint**.

THANK YOU

Tell us how it went



POST-WORKSHOP SURVEY

Your feedback shapes the next one — what landed, what dragged, what you want more of. Scan the code, or grab us over coffee. Thank you for spending two days on this with us.